

Muhammad Nabeed Haider

Computer Science Undergraduate

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in Python backend development, REST API design, and AI-powered applications. Passionate about building maintainable software, solving real-world problems, and writing clean, scalable code while continuously learning modern software engineering practices.

EDUCATION

Bachelor of Science in Computer Science

2026

FAST National University of Computer and Emerging Sciences, Islamabad

- CGPA: 3.56 / 4.00
- Relevant Coursework: Generative AI, Machine Learning, MLOps, Natural Language Processing, Statistical Modelling, Cloud Computing, Information Security

EXPERIENCE

AI Lab Demonstrator

Spring 2026

FAST National University of Computer and Emerging Sciences

- Conducted weekly Artificial Intelligence laboratory sessions for approximately 50 undergraduate students.
- Provided guidance on Python programming, search algorithms, machine learning fundamentals, and debugging techniques.
- Assisted with grading programming assignments, evaluating lab performance, and supporting students during practical sessions.
- Collaborated with the course instructor to maintain a consistent and engaging laboratory environment.

PROJECTS

GymBuddy

2026

Python • FastAPI • Ollama

- Developed the backend of a fitness management application using Python and FastAPI with a modular REST API architecture.
- Designed APIs for workout planning, exercise management, and user data handling.
- Collaborated within a development team using Git and version control practices to implement and integrate application features.
- Focused on writing maintainable backend code with clear project structure and scalable API design principles.

SCAR

2026

Python • LangGraph • FastAPI • Ollama

- Designed a modular multi-agent AI consulting platform using LangGraph and Python.
- Implemented specialized AI agents capable of task delegation and collaborative reasoning using locally hosted large language models through Ollama.

- Developed a scalable architecture emphasizing modularity, extensibility, and maintainability to support future enterprise AI workflows.
- Integrated FastAPI to expose agent capabilities through a RESTful backend suitable for production deployment.

Production MLOps Pipeline

2026

Python • FastAPI • Docker • GitHub Actions

- Built a production-ready machine learning deployment pipeline using FastAPI and Docker with a modular project architecture.
- Implemented automated CI workflows with GitHub Actions to validate code quality, execute tests, and streamline deployment.
- Integrated Prometheus metrics and Grafana dashboards to monitor application health, inference performance, and operational metrics.
- Designed a scalable deployment workflow following MLOps best practices for reproducibility, observability, and maintainability.

FakeLess

2025–2026

Python • PyTorch • SpeechBrain • HiFi-GAN

- Developed an audio deepfake prevention system that learns imperceptible perturbations to reduce the effectiveness of neural voice cloning models while preserving speech quality.
- Designed and trained deep learning models using PyTorch, leveraging SpeechBrain speaker embeddings and HiFi-GAN neural vocoders within the training pipeline.
- Prepared and processed the LibriSpeech dataset, implementing preprocessing, feature extraction, and automated training workflows for large-scale experimentation.
- Evaluated the system using speaker similarity and speech quality metrics to balance robustness against voice cloning with perceptual audio quality.

TECHNICAL SKILLS

Languages: Python, Java, SQL, C/C++

Backend: FastAPI, REST APIs

Databases: SQLite, MongoDB, PostgreSQL

Tools: Git, GitHub, Docker, Linux, VS Code

Concepts: Object-Oriented Programming, Data Structures, Algorithms, Software Engineering

ACHIEVEMENTS

- AI Lab Demonstrator for the Artificial Intelligence course at FAST-NUCES.
- Developed multiple end-to-end AI and machine learning projects spanning Generative AI, MLOps, and backend systems.
- Hands-on experience with production-oriented machine learning workflows, CI/CD, Docker, monitoring, and LLM-based applications.